

Qatar-2b

R-1421

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_210516 · created 2026-08-01T06:17:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459350.8089 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.028
Transit depth (Rp/Rs)^2	1.6 +/- 0.71 [%]
Orbital Inclination (inc)	89.29 +/- 1.61
Airmass coefficient 1 (a1)	0.953 +/- 0.024
Airmass coefficient 2 (a2)	0.0253 +/- 0.0086
Scatter in the residuals of the lightcurve fit is	2.54 %
Best Comparison Star	#3 - [611, 152]
Optimal Method	PSF photometry
Transit Duration (day)	0.0738 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.126 ± 0.028 vs 0.16327 (deviation 1.31σ)
Duration fit vs published	0.0738 d vs 0.0767 d (deviation 1.18σ)
Mid-time O-C	-2.9 min
Depth SNR	4.4
Residual scatter	2.54 %

Light curve

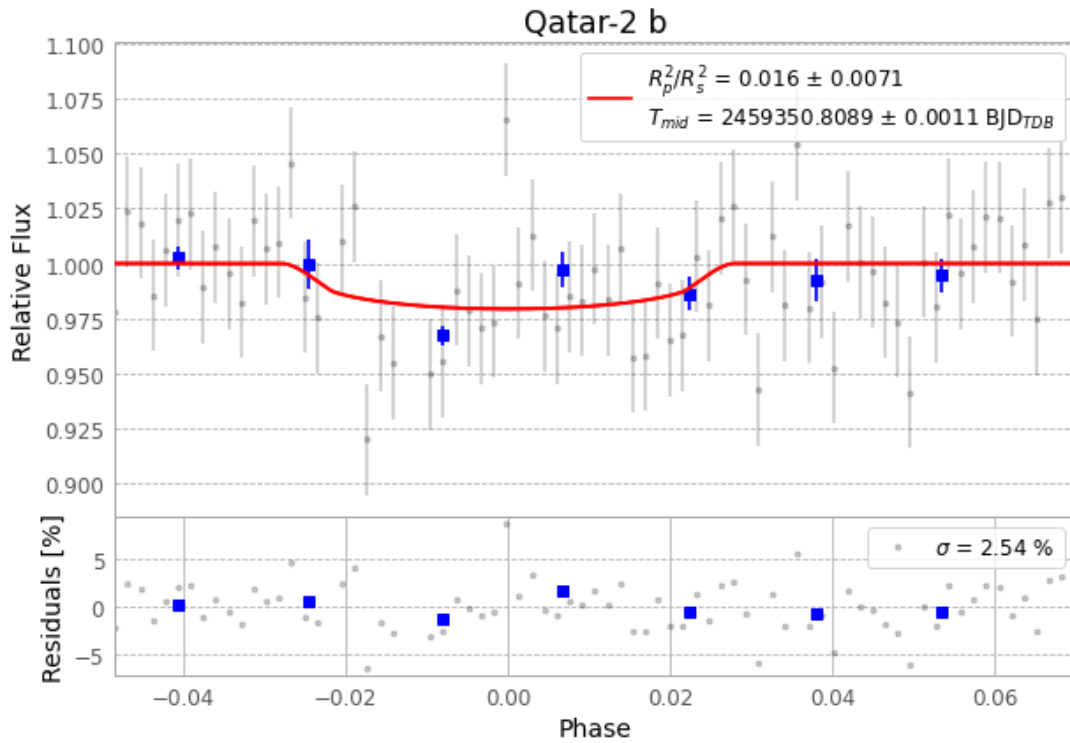


Figure 1 — FinalLightCurve_Qatar-2 b_2021-05-15.png (SHA-256 8497575e1fc280d2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [513, 28], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 52b740d65e74cf25...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), Qatar-2210516054212.FITS → Qatar-2210516093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-210516.log (4bf09e6004a1...), FinalLightCurve_Qatar-2 b_2021-05-15.png (8497575e1fc2...), FinalLightCurve_Qatar-2 b_2021-05-15.csv (2d4acd59b657...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 06:17Z.